

# Improving Nonlinear Modeling with Informative and Safe Gaussian Process Priors

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## Nonparametric flexibility might become excessive in the social sciences

**Small dataset sizes** cause unrealistic interpolation, extrapolation and violations of known constraints

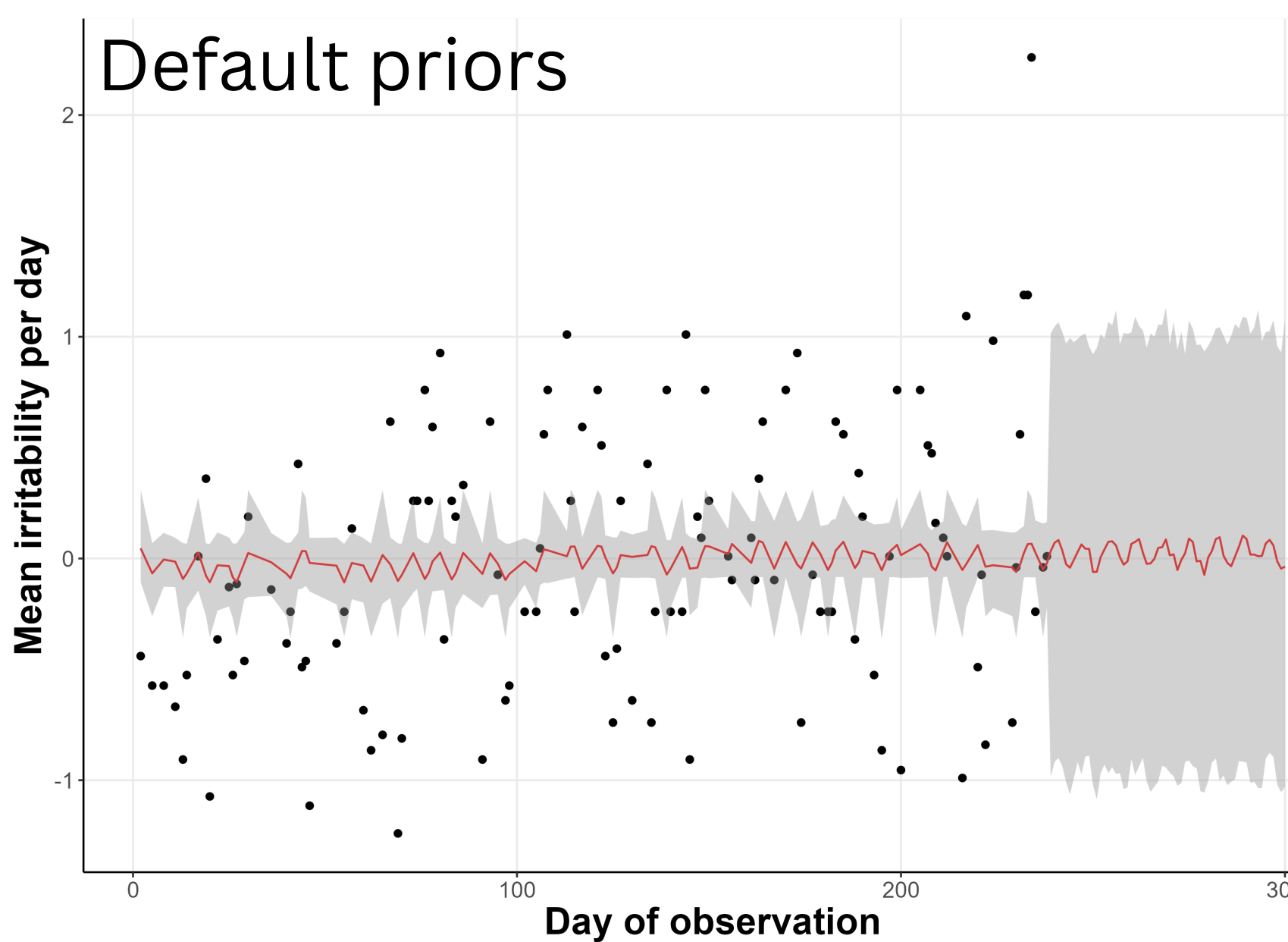


Domain and experts' knowledge exist **but are not straightforward to model**

## How we make informative GPs more accessible

- A paper focused on guidelines for the informative GP modelling and practical examples;
- Shiny app for the prior predictive sandbox-like visualization.

Default priors



### Informative hyperpriors

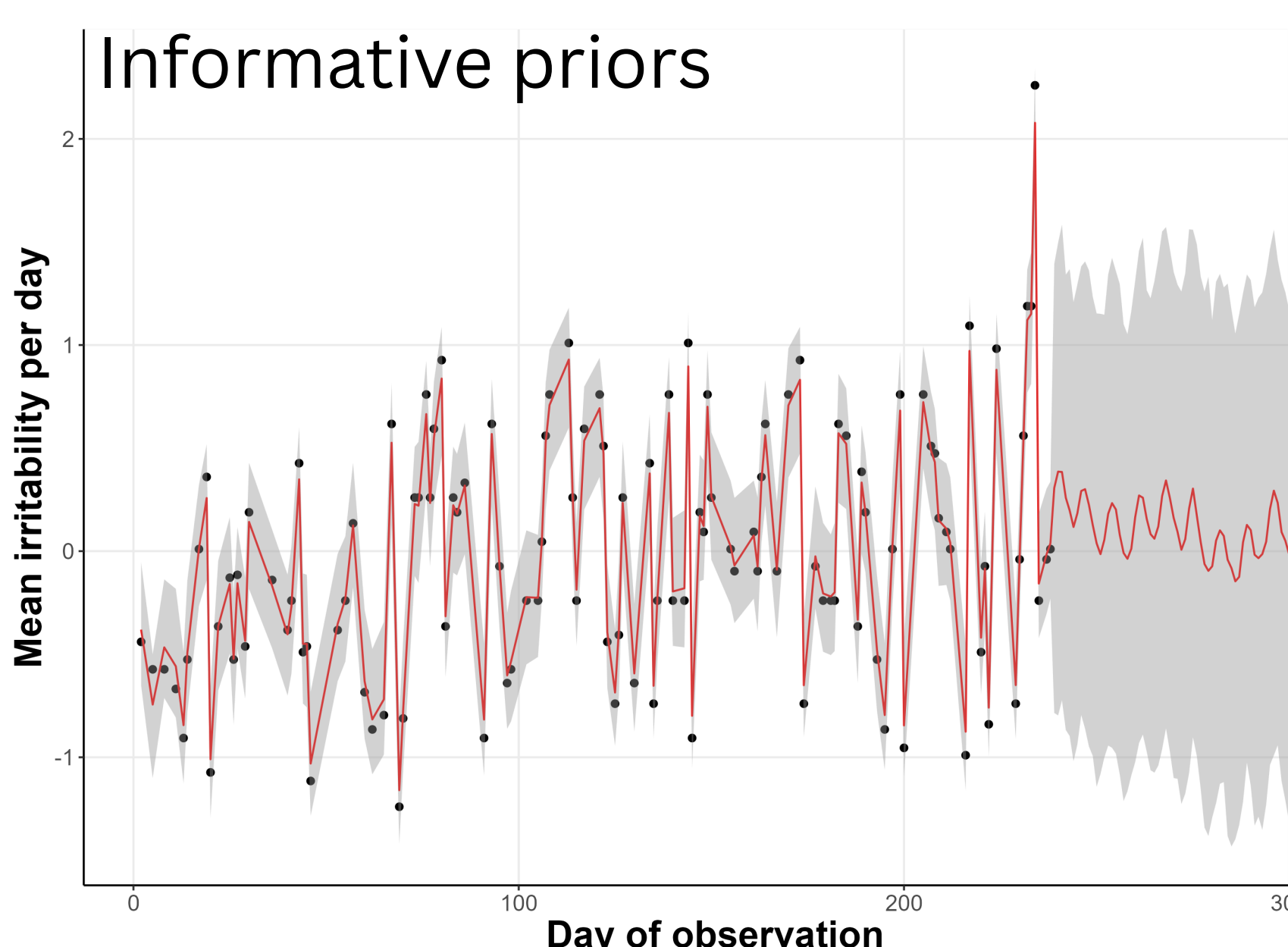
Data:

- ESM on irritability
- 1 participant
- 119 days

Model:

- priors and kernels based on theoretical expectations
- additive kernel composition
- rstan package

Informative priors



### Qualitative constraints

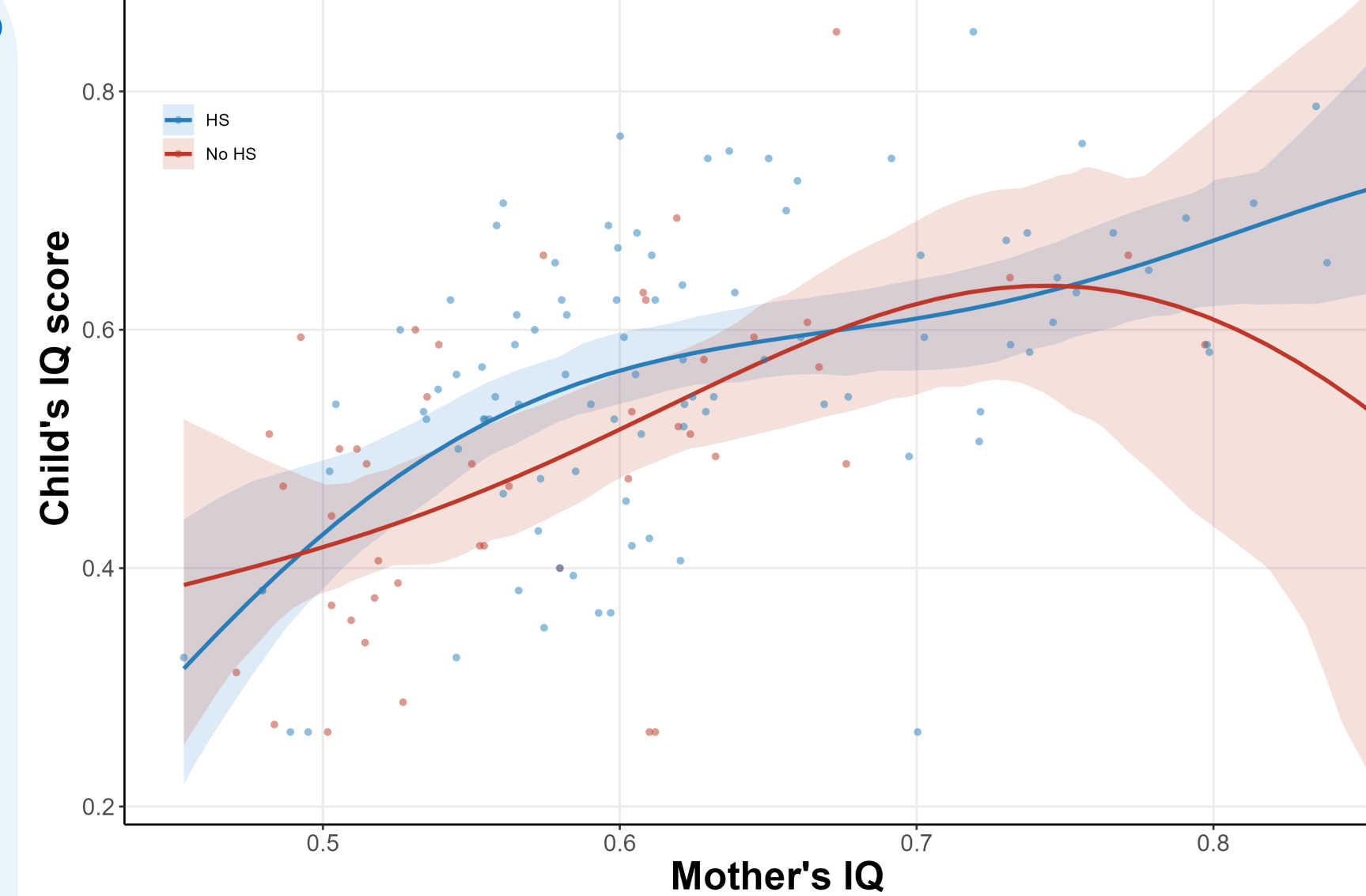
Data:

- Child's IQ based on mother's IQ and education level ("no hs" and "hs")
- 45 "no hs" and 100 "hs" participants

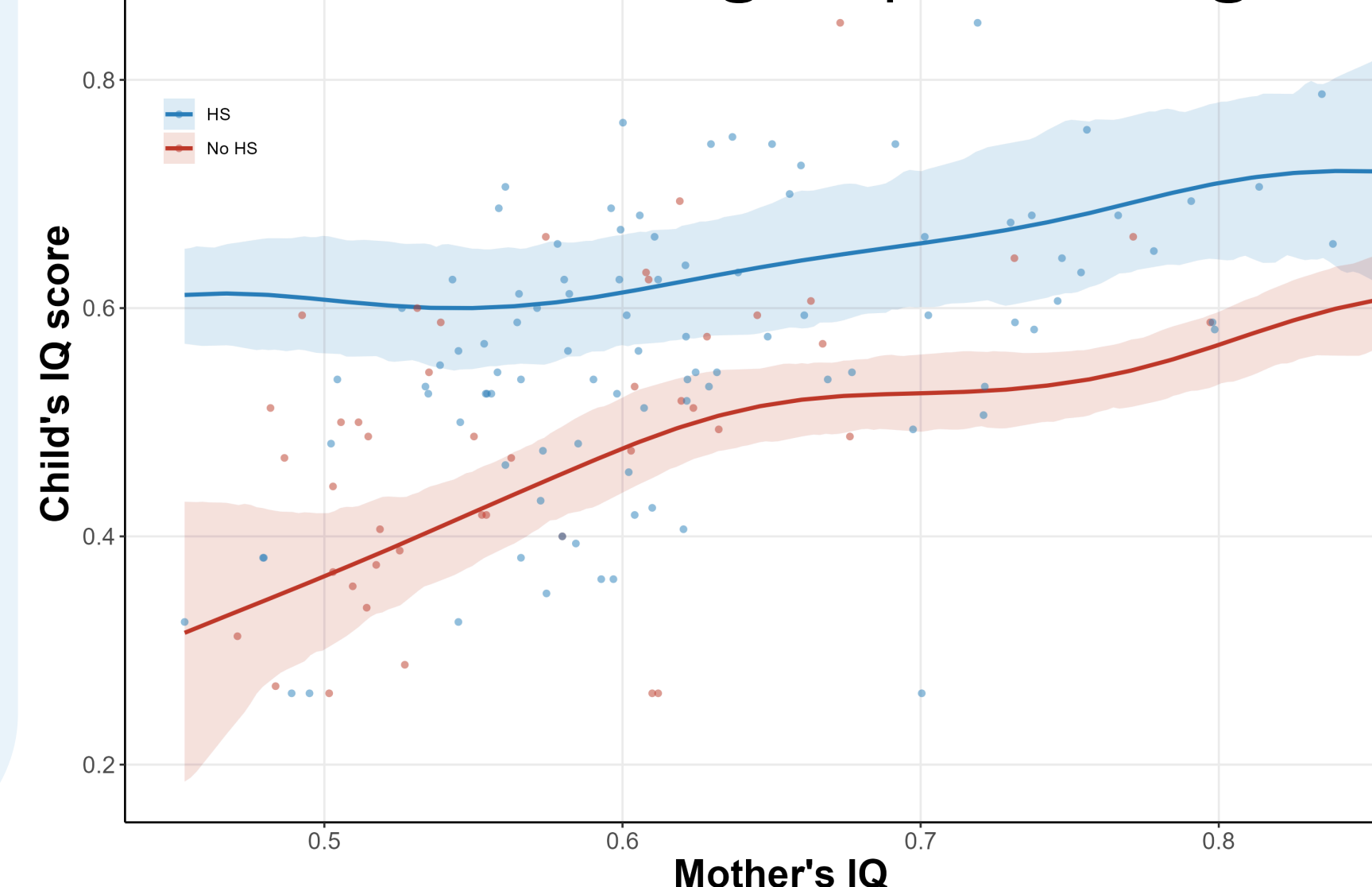
Model:

- y value bounds [0, 1]
- ordering of "hs" >= "no hs"
- lineggpr package

No constraints



Boundaries and group ordering



## Informative GPs

### Kernel choice and structure

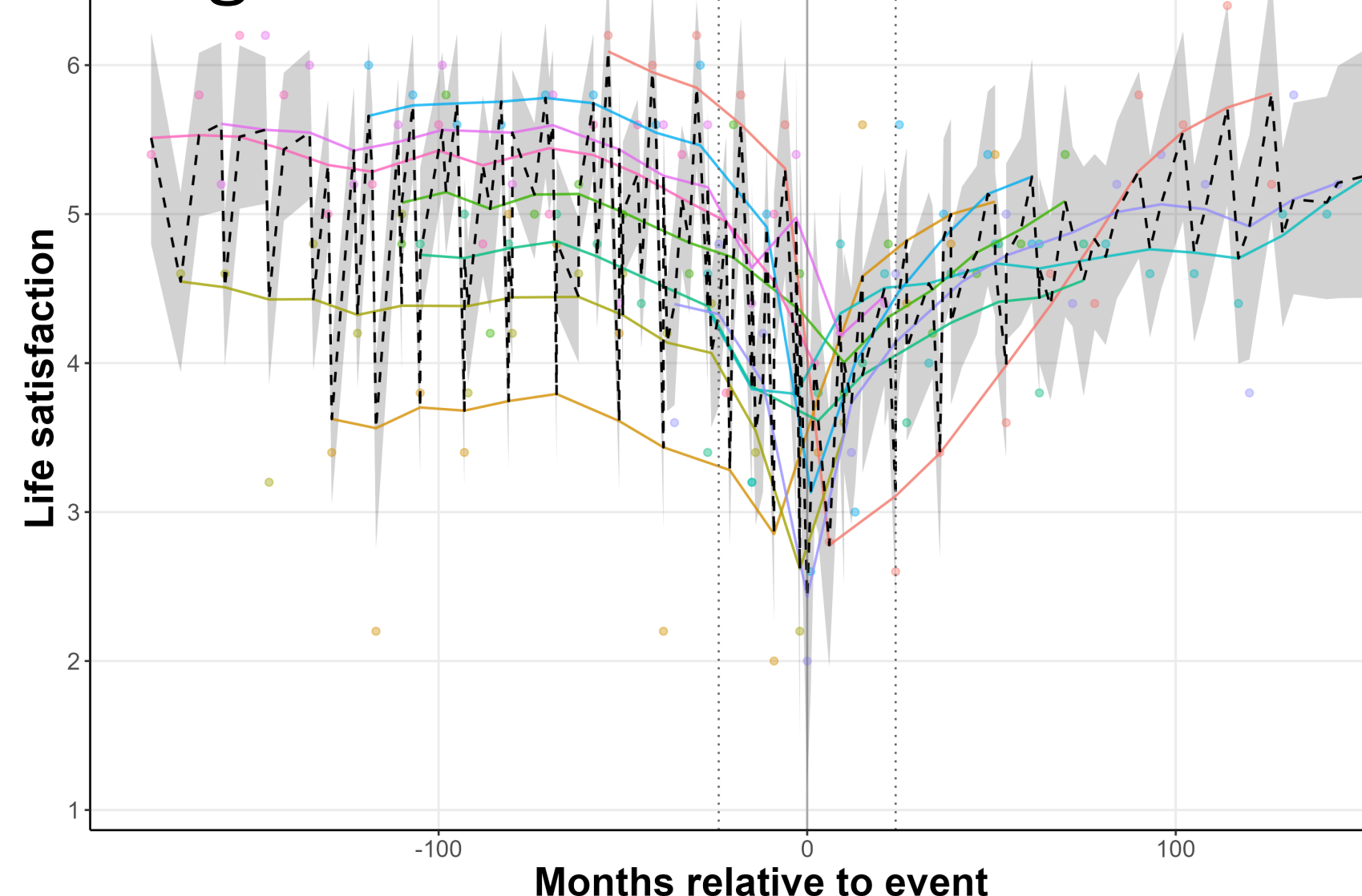
Data:

- Life satisfaction around widowhood
- 10 participants
- 15 observations each
- -178 to 153 months around the event

Model:

- changepoint kernel
- hierarchical
- priors based on theoretical expectations
- rstan package

Single "default" kernel



Changepoint kernel



input/output warping, etc...

### Prior predictive Shiny app

- illustrates how the choice of model parameters influences the prior;
- introduces the researcher to the key modelling points, in agreement with the notation of relevant R packages;
- Includes the choice of:
  - kernel structure (with changepoint and addition/multiplication combinations);
  - hyperpriors for kernel parameters;
  - bounds and monotonicity constraints.



## Forthcoming Research

- Strength of informativity of priors and their corresponding categories;
- Other ways to impose bounds;
- Qualitative restrictions that are most common and useful to the actual researchers.

## References



## Let us know what you think!



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